



Apple-Intel deal

Reportedly, Apple will be acquiring majority of Intel's smartphone modem business for \$1 billion. <https://dgit.in/aug19-59>



Huawei CEO's flex

The Huawei CEO, Richard Yu was seen using the unreleased Huawei Mate X, foldable phone. <https://dgit.in/aug19-60>

AMD Radeon RX 5700XT Graphics Card

A competitor, at last

AMD finally unveiled their Navi-based graphics cards at COMPUTEX 2019. AMD has

been the favourite when it comes to providing gamers value for their hard-earned money.

The Radeon RX 5700 XT and RX 5700 feature AMD's new RDNA architecture. A design that's primarily GCN with some elements of RDNA. Think of Navi as an augmented version of GCN with new Compute Units (CU), improved graphics pipeline and a multi-level cache hierarchy.

AMD Radeon RX 5700 XT / 5700 Specifications

Having moved to a 7nm manufacturing process, Navi allows for higher clocks while saving power. Thanks to smaller transistors, AMD managed to squeeze 10.3 billion transistors onto a die. The Compute Units still feature the same number of Stream Processors i.e. 64 but there have been other improvements.

The improvement with clock speeds are quite considerable. The peak boost now goes all the way up to 1905 MHz with the total board power scaling up to 225W as per the specs. The other major switch happens to be the memory type which is now GDDR6 instead of HBM2 which is cheaper but has reduced bandwidth per chip.

Performance

We start with the synthetic benchmarks within 3D Mark. The Radeon VII is still better than both new cards but not by a vast margin. The Radeon VII launched at \$699 whereas the RX 5700 XT costs \$399 and the RX 5700 costs \$349. So the new, cheaper cards coming this close shows how much more efficient the new Navi architecture is. Even in Time Spy and Time Spy Extreme, we see a similar pattern. Compared to the competition, the Radeon RX 5700 XT is better than the NVIDIA RTX 2070 and the RX 5700 scores lower than the 2070.



Price
₹36,568

Gaming

In Battlefield V, the RX 5700 XT did better than even the RTX 2080 across all tested resolutions which felt like an anomaly. On the other hand, the RX 5700 performed better than the Vega 64, RTX 2060 and even the older RX 590.

When it came to running Far Cry 5 and Far Cry New Dawn, the RX 5700 XT performed better than the RTX 2070 and even the older Vega 64. The RX 5700 equalled the RTX 2070 at 4K resolution but the differences became more apparent at lower resolutions as the RTX 2070 crept ahead. Even here, we see the RX 5700 coming ahead of the RTX 2060.

The Witcher 3 was the only game where the RTX 2070 performed better than the RX 5700 XT with the latest driver update. The RX 5700 XT did perform better with the previous driver that was initially seeded during the review process. Only at lower resolutions did the RX 5700 XT get the better of the RTX 2070. The same pattern ensued with the RX 5700 and the RTX 2060.

Overclocking

First, we tried the Auto OC settings within WattMan and we tried messing around with the OC manually to extract whatever we could. Here are the results for the 5700 XT while we continue to work on the 5700:

Individually, the GPU and the Mem clock were bumped up to find out their



FEATURES.....	79
PERFORMANCE.....	80
VALUE.....	75
DESIGN.....	78

peak values. The base clock could go all the way up to 2130 MHz whereas the memory could go up till 935 MHz. With both parameters set to their peak values we couldn't get a proper run and they would crash every single time. We then lowered the base clock all the way down to 1950 MHz but

the benchmark would still crash. So we set the base clock to 2100 and then lowered the memory clock. At 930 MHz memory clock, the OS stopped crashing but the benchmark was still failing. So after dropping the memory clock to 925 MHz, we finally ended up with the first successful run of Time Spy with a score of 9598. Compared to the highest score we've obtained thus far at stock clocks, which is 9193, this OC attempt got us a 4.41% improvement in performance.

Verdict

AMD's Navi is not only performing better than competing solutions but is also better priced with the AMD Radeon RX 5700 XT priced at Rs.30,990+Taxes and the RX 5700 priced at Rs.26,990+Taxes. These cards displace the RTX 2070 and the RTX 2060 from the sweet spot, and with their new RDNA architecture carve a good share of the GPU market for now.

—Mithun Mohandas

SPECIFICATIONS

STREAM PROCESSORS: 2560 | CU: 40 | BOOST CLOCK: 1905 MHz | VRAM: 8 GB GDDR6 | BUS-WIDTH: 256-bit | CONNECTORS: 1x8PIN+1x6-PIN | TBP: <225W

CONTACT

AMD (ADVANCED MICRO DEVICES) | PHONE: Web form | EMAIL: Web form | WEBSITE: www.amd.com